

Transformer u(t) Experiment Report

Manuscript time-dependent control trained by PMP/KKT optimality gaps

This report reproduces the manuscript's time-dependent Transformer control strategy $u_{\text{theta}}(t)$: $[0,T] \rightarrow [0,u_{\text{max}}]$.

1. Model and Training Objective

We use the population dynamics and cost from the manuscript:

$$\dot{N}_i(t) = (r_i - \phi_i u(t) - M_i G(N(t))) N_i(t)$$

$$G(N) = \log \left(1 + \frac{1}{m} \sum_{k=1}^m N_k \right)$$

$$J(u) = \alpha^\top N(T) + \int_0^T (\beta^\top N(t) + \gamma u(t)) dt$$

Reported parameters: $T=10$, $m=21$, $u_{\text{max}}=3$, $\alpha=1$, $\beta=0.1$, $\gamma=20$, and $N_i(0)=10$. The control is represented by a small Transformer encoder over normalized time: $u_{\text{theta}}(t_k)=u_{\text{max}} \text{sigma}(g_{\text{theta}}(t_k))$.

Given $u_{\text{theta}}(t)$, we roll out $N_{\text{theta}}(t)$, solve the costate equation backward, and compute the switching function

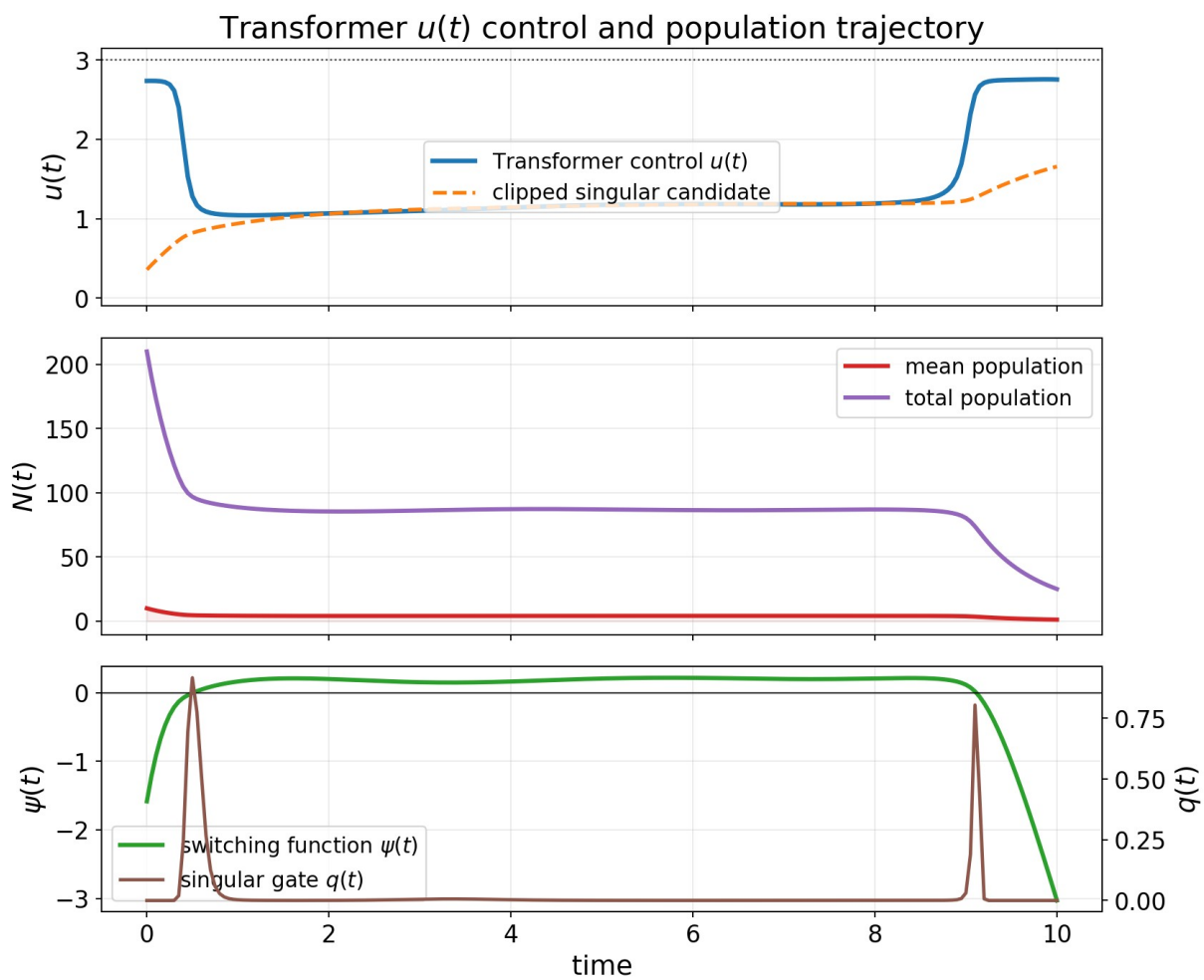
$$\psi(t) = H_u(N, \lambda, u) = \gamma - \sum_i \phi_i \lambda_i(t) N_i(t)$$

component	role
non-singular KKT loss	enforces $u=0$ when $\psi>0$ and $u=u_{\text{max}}$ when $\psi<0$
singular loss	near $\psi=0$, matches $u_{\text{theta}}(t)$ to the singular candidate $u_{\text{sing}}(N(t))$

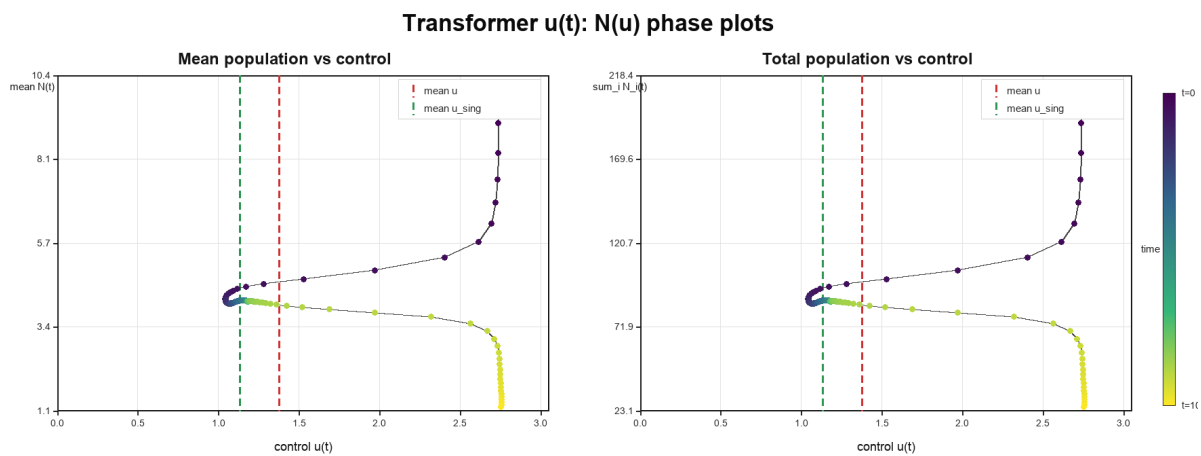
Here $q(t)$ is a smooth weight that switches between the singular condition near $\psi(t)=0$ and the boundary KKT condition away from $\psi(t)=0$.

2. Learned u(t), N(t), and N(u)

The learned control starts high, transitions to a lower interior/singular-like region, and increases again near the terminal portion.



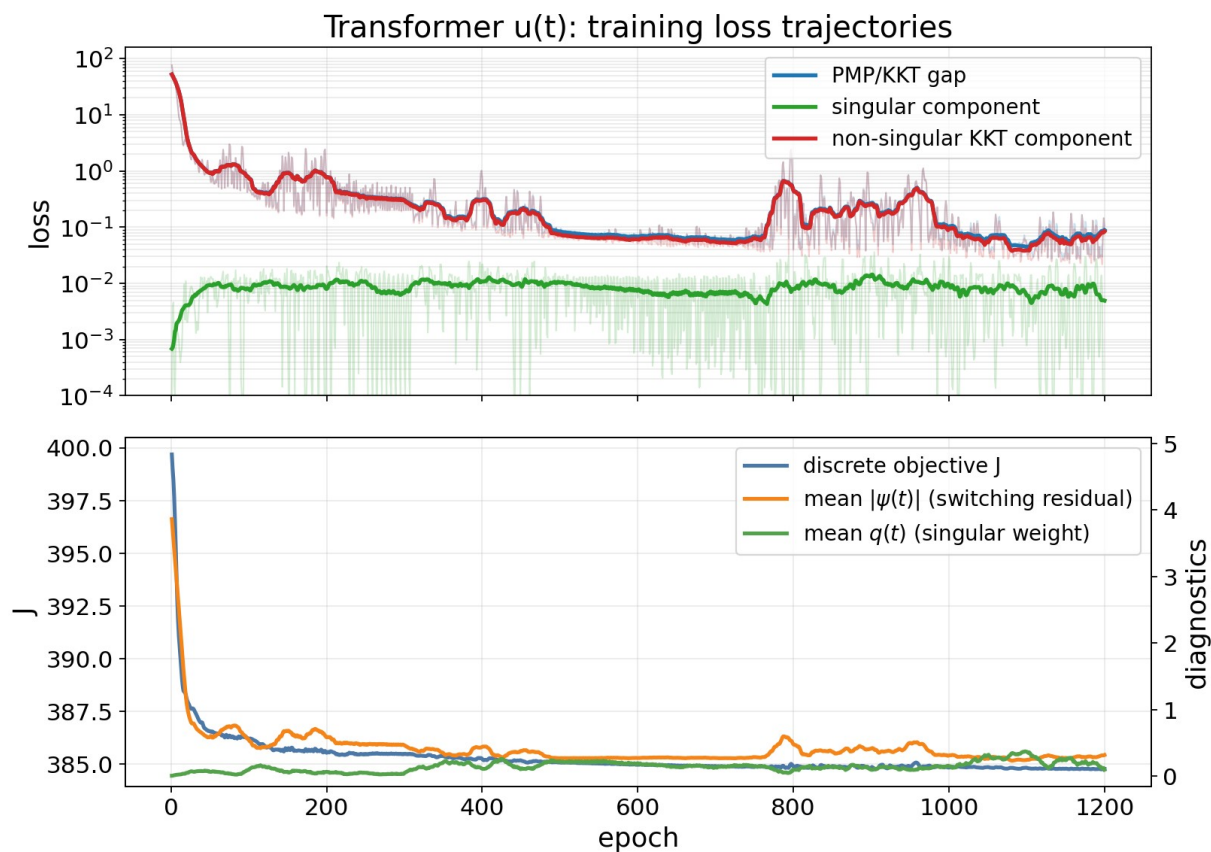
The $N(u)$ phase plot shows population directly against the applied control value; color indicates time.



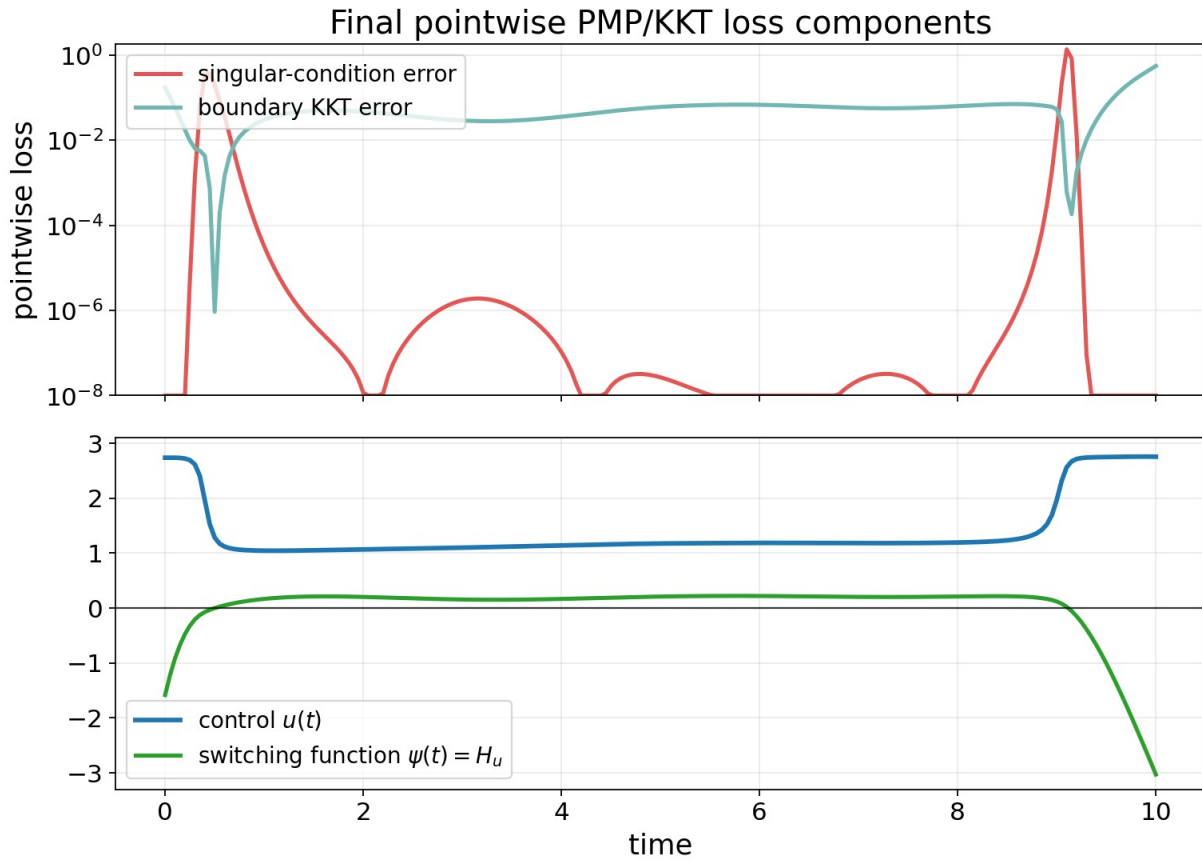
3. Training Loss Trajectories for PMP/KKT Conditions

The table reports the smallest recorded training loss. In this experiment, the training loss is the manuscript's PMP/KKT optimality gap, composed of the singular condition and the non-singular Hamiltonian minimization condition.

metric	value
total training loss / PMP-KKT optimality gap	0.02637
singular-condition training loss	0.00167
non-singular Hamiltonian-minimization training loss	0.02471
objective J on the training grid	384.76
range and mean of $u_{\text{theta}}(t)$	min 1.038, max 2.758, mean 1.372
terminal mean state	1.207



The final pointwise diagnostic below uses a smooth weight $q(t)$ to separate the two regimes: near $\psi(t)=0$ it emphasizes the singular-condition error; away from $\psi(t)=0$ it emphasizes the boundary KKT error.



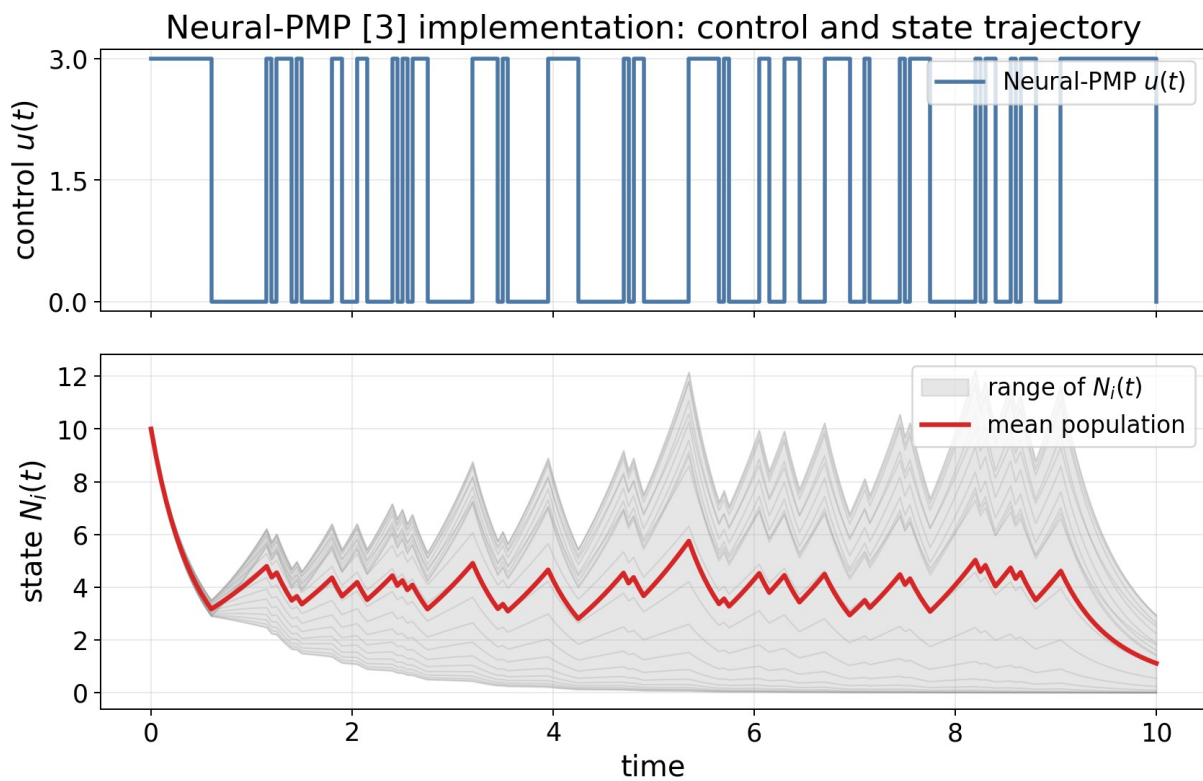
4. Comparison With Related Work [3]

Related work [3], Neural-PMP / PMP-gradient, also follows a Pontryagin-style procedure: forward state integration, backward costate recursion, and Hamiltonian-gradient updates of a discrete control sequence. For comparison on the same model and parameters, we implemented the corresponding control-update step from [3].

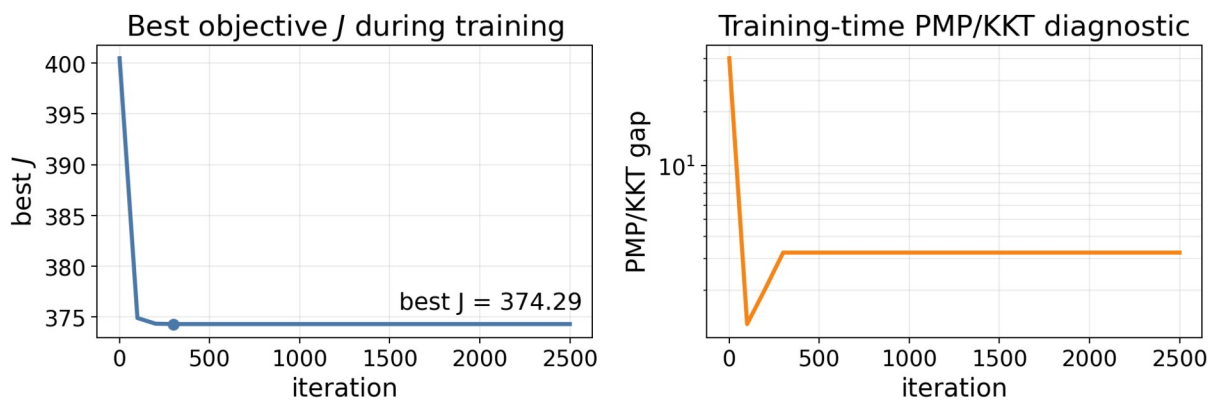
method	control update / training criterion	PMP/KKT gap	objective J^*	note
Transformer $u(t)$	paper PMP/KKT optimality-gap loss	0.0523	386.70	main $u(t)$ reproduction
Neural-PMP [3]	Hamiltonian-gradient update of control sequence	7.47	387.02	related-work comparison
direct grid cost	minimize discretized J	0.805	386.47	reference only
constant $u=1.5$	no training	76.89	400.40	scale check

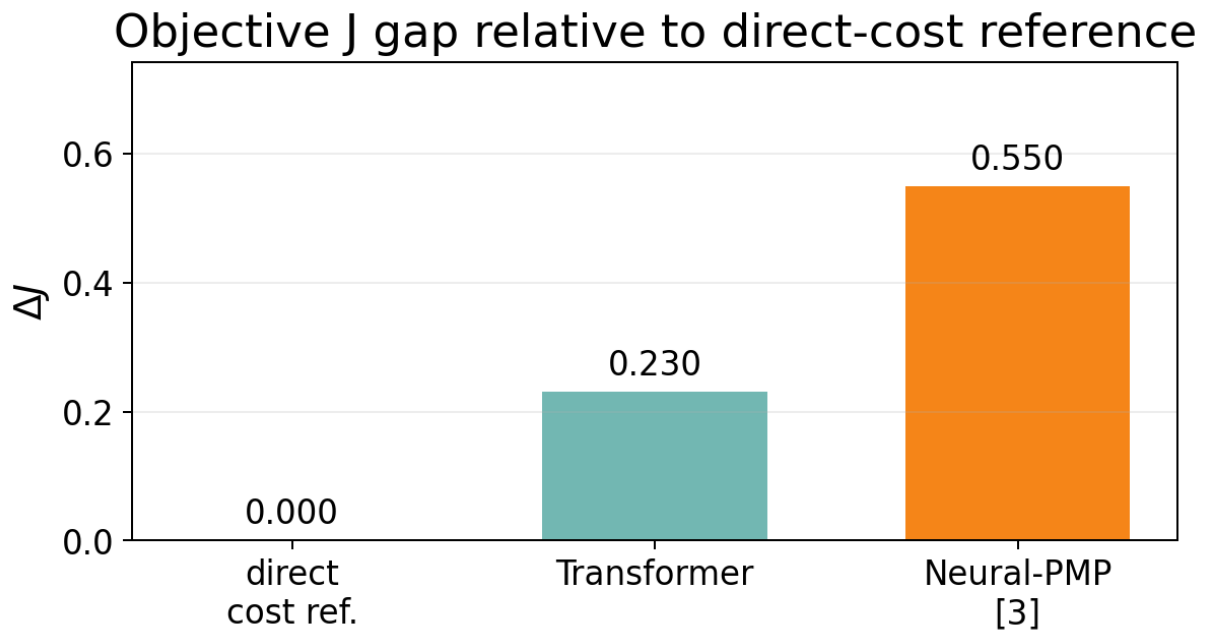
* The J values in this table are computed after fixing $u(t)$, reintegrating $N(t)$ with a finer-step fourth-order Runge-Kutta method, and then applying the manuscript objective definition. This is only to use the same numerical integration accuracy across methods.

Under the same PMP/KKT gap calculation, the Transformer $u_{\theta}(t)$ has a much smaller gap than the Neural-PMP implementation of [3]. When J is recomputed with the same numerical evaluator, the Transformer also has a slightly lower objective value in this run.



Neural-PMP [3] implementation: selected-run training trajectories





5. Conclusion

The requested $u(t)$ reproduction is complete. The Transformer control trajectory trained with the manuscript's PMP/KKT optimality-gap loss is smooth and satisfies the control bounds, reduces the training optimality gap from 76.26 to 0.02637, and gives J approximately 386.70 under the common numerical evaluation. The state-dependent extension $u(t, N)$ is not included in this report because it requires different optimality conditions.